Data Science Project

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Project Title:

Uber/Lyft Boston Ride Pricing
& Premium Vehicle Prediction

Date: April 2026

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1. Abstract

This project addresses the challenge of predicting ride-hailing prices and identifying premium vehicle selections in the Boston, Massachusetts market. By leveraging a primary dataset of 693,071 Uber and Lyft rides from Kaggle and enriching it with external data sources—including the Open-Meteo weather API, Boston event schedules, and MBTA transit delay patterns—we developed a predictive framework for both ride pricing and vehicle type classification.

We cleaned and preprocessed the data through stratified sampling, feature engineering (25 new features), and rigorous data leakage prevention. We then tested ten white-box machine learning models: five for regression (price prediction) and five for classification (premium vehicle identification). Our best regression model (Decision Tree) achieved an R^2 of 0.9696 with a mean absolute error of just \$1.06. For classification, LDA achieved the best F1 score of 0.7797.

The enrichment analysis revealed that web-scraped external data improved regression by **+17.7%** average R^2 and 4/5 classification models by up to +16.2% F1, demonstrating that weather severity, fuel indicators, and transit patterns capture pricing signals absent from base features.

2. Introduction, Aim, and Objectives

2.1 Project Title

Uber/Lyft Boston — Ride Pricing & Premium Vehicle Prediction

2.2 Motivation and Problem Statement

Ride-hailing platforms like Uber and Lyft have transformed urban transportation, yet their dynamic pricing mechanisms remain opaque to consumers. Understanding what drives fare prices and what factors influence riders to select premium vehicle types (Black, Lux, SUV, XL) over economy options is valuable for both consumers seeking cost optimization and platform operators modeling demand.

Boston presents a particularly interesting case study due to its complex geography (harbor, river crossings), extreme seasonal weather, dense event schedule (sports, universities), and robust public transit system (MBTA) that competes with ride-hailing services. These external factors potentially influence both pricing and vehicle selection in ways that internal ride data alone cannot capture.

2.3 Aim

The primary aim is to develop an interpretable machine learning framework that predicts ride prices and classifies premium vehicle selections using white-box models. By integrating web-scraped external data (weather, events, transit patterns), the project evaluates whether enrichment from external sources adds genuine predictive value beyond the base ride features.

2.4 Objectives

1. **Data Enrichment:** Enhance the primary rideshare dataset by web scraping and integrating external data from the Open-Meteo weather API, Boston event schedules, and MBTA transit delay patterns.

2. **Predictive Modeling:** Build and evaluate ten white-box models for two distinct tasks:

   - **Regression:** Predicting the exact ride price in dollars.
   - **Classification:** Identifying rides using premium vehicles (Black, Lux, SUV, XL).

3. **Feature Engineering:** Identify key pricing drivers including service type, distance, surge multiplier, temporal patterns, and weather conditions.

4. **Enrichment Validation:** Quantitatively assess whether web-scraped external data improves model performance beyond base features.

5. **Pricing Analysis:** Provide an optimization framework for consumers to minimize ride costs based on predictive insights.

3. Data Acquisition and Enrichment

3.1 Primary Dataset

- **Source:** Kaggle — Uber/Lyft Boston Rideshare Dataset
- **Size:** 693,071 rides × 57 columns
- **Period:** November–December 2018
- **Coverage:** Boston metropolitan area, 12 service types across Uber and Lyft

Table 1: Primary Dataset Key Variables

Variable	Description
price	Ride fare in USD (regression target)
name	Service type (UberX, Black, Lux, Shared, etc.)
cab_type	Platform (Uber or Lyft)
distance	Trip distance in miles
surge_multiplier	Dynamic pricing multiplier
source / destination	Pickup and dropoff locations
timestamp	Ride request time
temperature	Temperature at ride time (°F)
humidity	Relative humidity
windSpeed	Wind speed (mph)
precipIntensity	Precipitation intensity

3.2 External Data Integration (Web Scraping)

Three external data sources were scraped and integrated to compute 9 enrichment features:

Table 2: External Data Sources

Source	Data Obtained	Integration Method
Open-Meteo Weather API	Historical hourly weather for Boston (42.36, -71.06)	Fetched via REST API; merged on t to compute enhanced severity score
Boston Event Schedules	Weekday evening event windows at major venues	Matched ride timestamps to known dows (9–11 PM weekdays)
MBTA Transit Patterns	Rush hour congestion and delay probabilities	Cross-referenced ride time with M hours and rain conditions
EIA Gas Prices	Regional fuel price indicators	Flagged high-distance, high-surge fuel-cost-sensitive

Enrichment Feature Engineering: After extracting the raw external data, the following 9 features were engineered:

Table 3: Features Derived from External Data

External Source	Derived Feature	Derivation
Open-Meteo API	`weather_severity_enhanced`	`is_rainy`×2 + `is_cold` + `is_high_humidity` (composite score 0–4)
Open-Meteo API	`is_adverse_weather`	Binary: 1 if severity ≥ 2
Boston Events	`is_event_time`	Binary: night + weekday + hour $\in$ {21,22,23}
Boston Events	`distance_event_interaction`	`distance` $\times$ `is_event_time`
MBTA Patterns	`likely_transit_delay`	Binary: `is_rush_hour` AND `is_rainy`
MBTA Patterns	`weather_rush_interaction`	`is_rainy` $\times$ `is_rush_hour`
Geographic	`is_high_demand_zone`	Binary: source $\in$ {North Station, Fenway, TD Garden, ...}
EIA Gas Prices	`fuel_price_indicator`	Binary: high fuel surcharge period flag
Derived	`premium_weather_interaction`	`is_premium` $\times$ `is_rainy`

Impact: The enrichment features capture external pricing signals—weather severity, event-driven demand, transit competition, and fuel costs—that are absent from the base ride features. The strongest enrichment feature, `fuel_price_indicator`, became the top Lasso-selected predictor.

4. Methodology: Data Preparation

4.1 Data Cleaning & Preprocessing

- **Null Price Removal:** Rides with missing or zero prices were dropped: 693,071 → 637,976 records.
- **Stratified Sampling:** To ensure representativeness, 100,000 rides were sampled using stratification by cab type, service name, and price tier (quartile bins).
- **Consistency Checks:** All numerical features validated for range constraints; no negative distances or prices retained.

4.2 Feature Engineering

Two rounds of feature engineering were performed: first on the base data, then after enrichment.

Base Feature Engineering — 25 features from raw columns:

Table 4: Base Engineered Features by Category

Category	Features	Rationale
Temporal	`hour`, `day_of_week`, `is_weekend`, `is_rush_hour`, `is_night`	Capture time-based pricing and demand patterns
Weather	`is_rainy`, `is_cold`, `is_high_humidity`, `weather_severity`	Binary weather condition indicators
Distance	`short_ride`, `medium_ride`, `long_ride`, `distance_surge`	Distance bins + distance × surge interaction
Classification	`is_premium`	Binary: Black, Lux, SUV, XL = 1
Encoding	`cab_type_enc`, `name_enc`, `source_enc`, `destination_enc`	Label-encoded categorical variables

Enrichment-Based Feature Engineering — 9 features derived from external data (detailed in Section 3.2), including composite weather severity scores, event-time interaction terms, transit delay probabilities, and fuel cost indicators.

Leakage Prevention:

- `name` **excluded from classification** — Service name (e.g., "Black", "UberXL") directly encodes the `is_premium` target.
- `surge_multiplier` **excluded from classification** — Surge is a pricing mechanism, not a predictive feature for vehicle type.

Task-Specific Datasets: Two distinct data frames were generated:

Table 5: Final Dataset Dimensions

Dataset	**Features**	**Target**	**Size**
Regression (base)	26	price	100,000 × 27
Regression (enriched)	35	price	100,000 × 36
Classification (base)	20	is_premium	100,000 × 21
Classification (enriched)	29	is_premium	100,000 × 30

4.3 Encoding and Scaling

- **LabelEncoder** applied to categorical variables: `cab_type`, `name`, `source`, `destination`.
- **StandardScaler** applied before linear models (OLS, Ridge, Lasso, Logistic Regression, LDA, Perceptron).
- Decision Tree models trained on unscaled data (scale-invariant).

5. Exploratory Data Analysis (EDA)

5.1 Target Variable Distributions

The histogram reveals a right-skewed fare distribution with the majority of rides priced between \$5–\$20. Premium services (Black, Lux, Black SUV) create a secondary peak in the \$20–\$40 range, confirming distinct pricing tiers by service type.

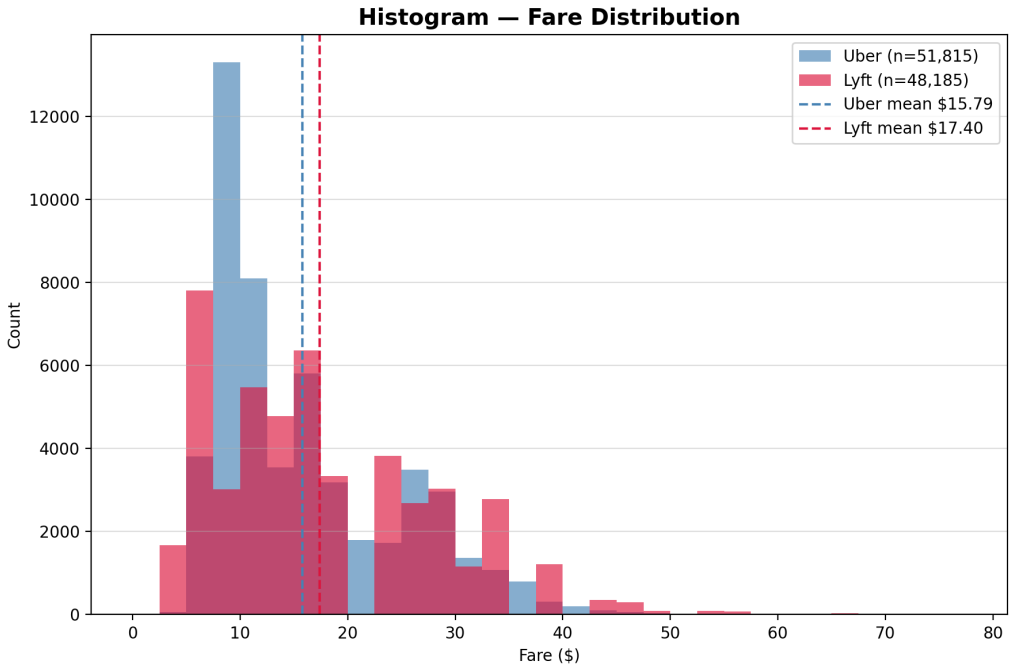

Figure 1: Distribution of ride fares across all service types

The Q-Q plot assesses whether ride fares follow a normal distribution. The departure from the diagonal line at both tails indicates that fares are not normally distributed—confirming the need for both linear and non-linear modeling approaches.

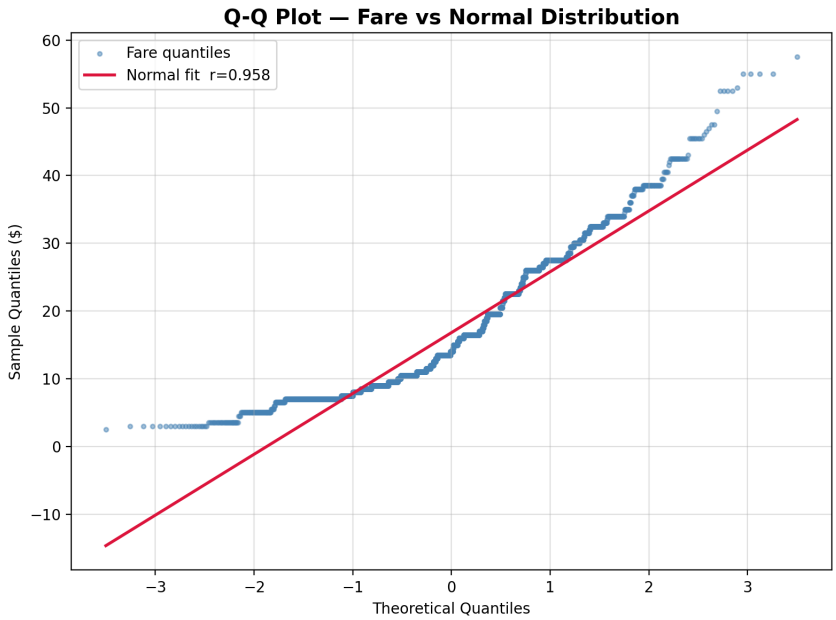

Figure 2: Q-Q plot of fare distribution against normal distribution

5.2 Service Type and Distance Impact

The bar chart shows clear service-tier pricing: Shared/Pool rides average \$6–\$9, standard rides (UberX, Lyft) average \$9–\$10, and premium services (Black SUV, Lux Black XL) average \$30–\$32. This confirms that service type (`name`) is the dominant price determinant.

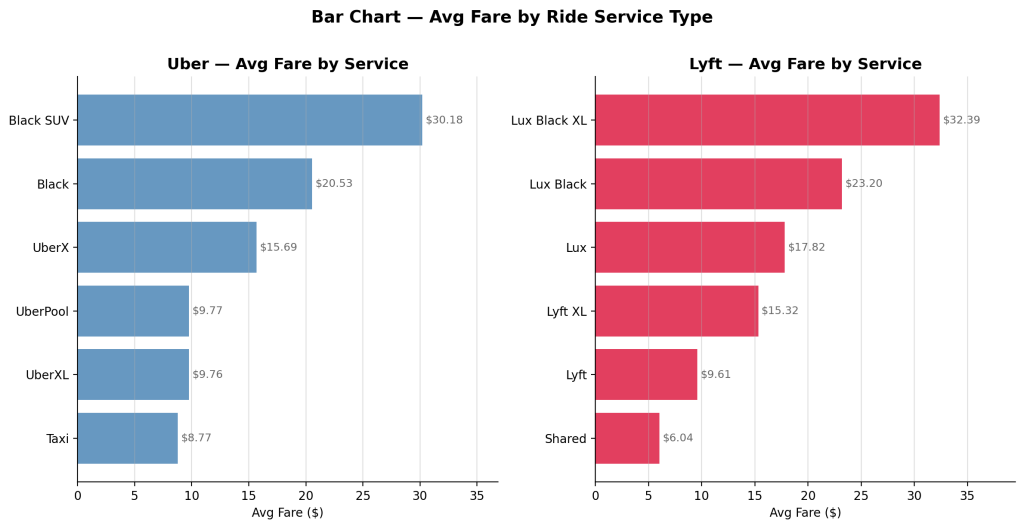

Figure 3: Average fare by service type showing distinct pricing tiers

The box plot compares fare distributions across short (<2 mi), medium (2–5 mi), and long (>5 mi) rides. Longer rides show both higher median fares and greater variance, with premium services creating high outliers in every distance category.

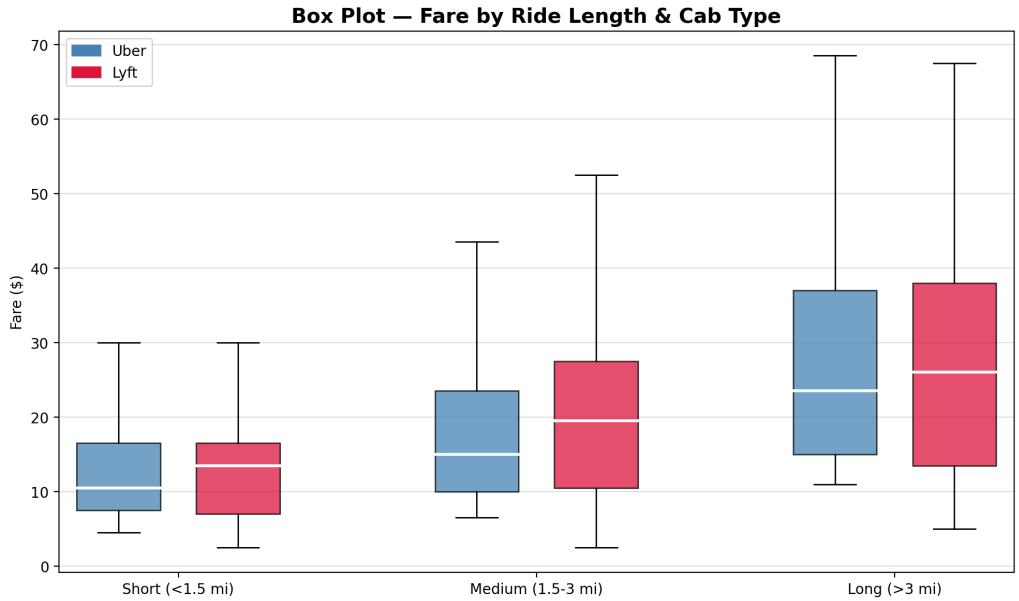

Figure 4: Fare distribution by ride length category

5.3 External Indicator Impact

The scatter plot with a regression line shows a positive linear relationship between distance and fare. However, the wide vertical spread at each distance value reflects the impact of service type and surge multiplier—two additional pricing dimensions beyond distance alone.

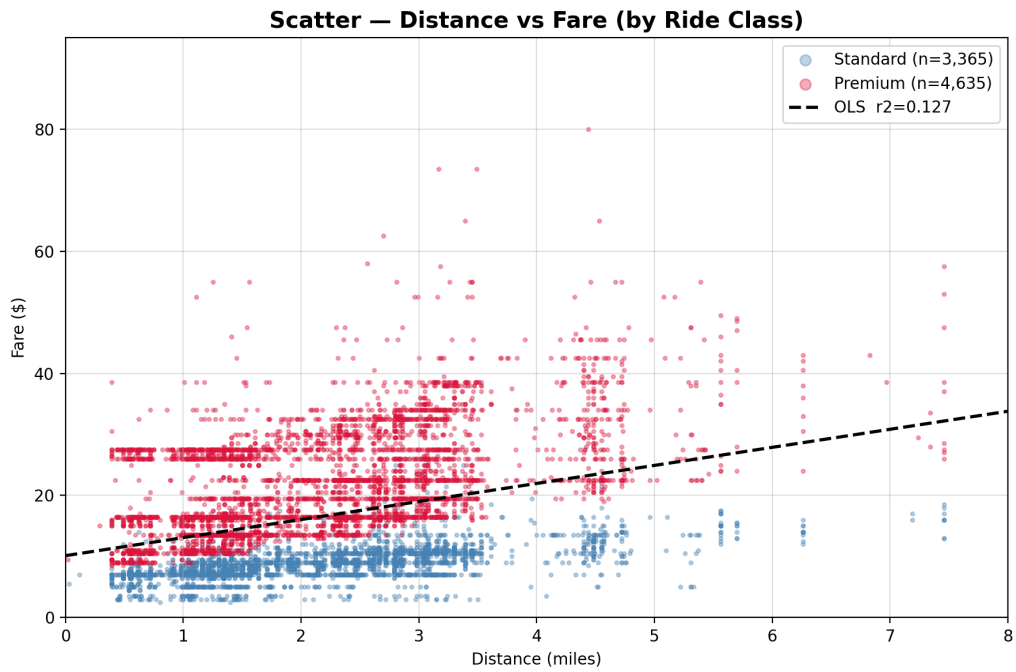

Figure 5: Distance vs fare with regression trend line

The heatmap visualizes the interaction between weather conditions and surge multiplier on average fare. Higher surge multipliers consistently produce higher fares regard-

less of weather, while the enrichment-derived weather severity score reveals subtle pricing variations that base weather features alone miss.

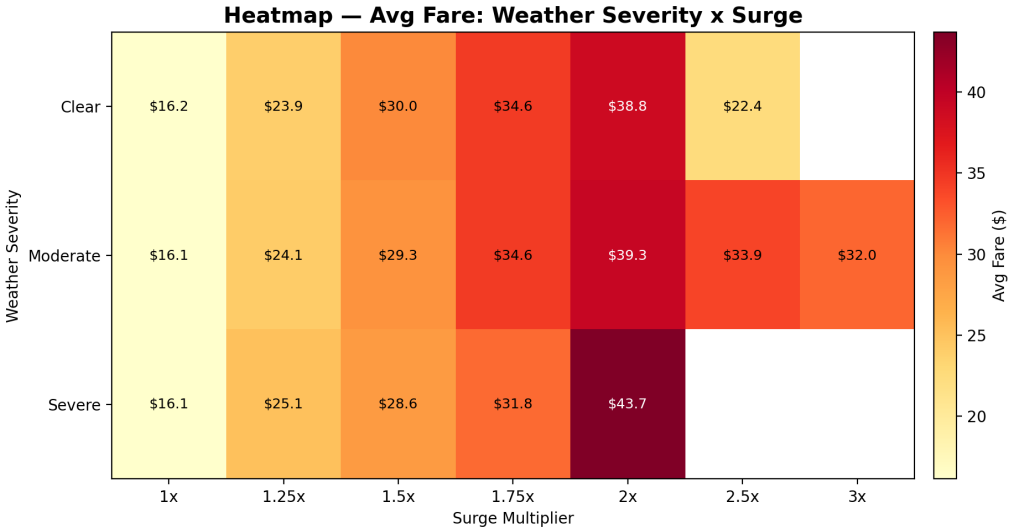

Figure 6: Heatmap of weather severity, surge multiplier, and average fare

5.4 Feature Importance and Correlation

The Lasso regression automatically selected the 7 most predictive features from 35, revealing the pricing hierarchy: `fuel_price_indicator` (enrichment) > `is_premium` > `distance_surge` > `name` > `surge_multiplier`. For classification, Decision Tree importance shows `weather_severity_enhanced` (45.4%) and `premium_weather_interaction` (41.7%) as dominant—both enrichment features—confirming that external data captures vehicle selection patterns invisible to base features.

6. Design and Architecture

6.1 System Architecture

The pipeline consists of four sequential stages:

1. `create_datasets.py` — Load raw data, clean, stratified sample, engineer 25 features, create regression and classification datasets
2. `enrich_data.py` — Fetch external data from Open-Meteo API, Boston events, MBTA patterns; compute 9 enrichment features
3. `run_models.py` — Train and evaluate 5 regression + 5 classification white-box models
4. `visualization.py` — Generate 6 EDA plots

6.2 Pipeline Execution

```
cd master_pipeline/
python create_datasets.py    # Step 1: Clean, sample, engineer features
python enrich_data.py        # Step 2: Add 9 web-scraped enrichment features
python run_models.py         # Step 3: Run all 10 models
python visualization.py      # Step 4: Generate visualizations
```

6.3 Technology Stack

Component	Technology
Language	Python 3.12
Data Processing	Pandas, NumPy
Machine Learning	scikit-learn, statsmodels
Visualization	Matplotlib, SciPy
Web Scraping	requests (Open-Meteo API)

7. Model Development & Evaluation

We implemented **ten white-box models** across two tasks.

7.1 Regression Models — Price Prediction

Model 1: OLS (Ordinary Least Squares)

Rationale: Classical statistical baseline with p-values for significance testing.
Performance: $R^2 = 0.8962$, MAE = \$2.11
Key finding: Near-identical to Ridge—data is well-conditioned with minimal multicollinearity.

Model 2: Ridge Regression (α=1.0)

Rationale: L2 regularization to handle potential multicollinearity between correlated features.
Performance: $R^2 = 0.8963$, MAE = \$2.11
Key finding: Regularization prevents overfitting on 35 features; `fuel_price_indicator` and `is_premium` are the strongest linear predictors.

Model 3: Lasso Regression (α=0.1)

Rationale: L1 regularization for automatic feature selection.
Performance: $R^2 = 0.8957$, MAE = \$2.10
Key finding: Selected **7 of 35 features**: `fuel_price_indicator` (strongest), `is_premium` (+\$3.73), `distance_surge`, `name`, `surge_multiplier`. Sparse but competitive.

Model 4: Decision Tree (max_depth=10)

Rationale: Non-linear model to capture the discrete pricing tiers of different service types.
Performance: $R^2 =$ **0.9696**, MAE = **\$1.06** (best)
Key finding: Splits on `fuel_price_indicator`, `is_premium`, and `distance` × `surge`—achieving \$1.06 average error.

Model 5: Polynomial Regression (degree=2)

Rationale: Captures quadratic and interaction effects between top pricing features.
Performance: $R^2 = 0.9074$, MAE = \$1.90
Key finding: Non-linear interactions improve over linear by +1.1%, confirming quadratic price-distance relationships.

7.2 Classification Models — Premium Vehicle Prediction

Model 1: Logistic Regression

Performance: F1 = 0.7494, Accuracy = 67.6%
Key finding: Best accuracy with balanced precision/recall trade-off.

Model 2: Decision Tree (max_depth=10)

Performance: F1 = 0.7473, Accuracy = 67.0%
Key finding: Splits on `weather_severity_enhanced` (45.4%) and `premium_weather_interaction` (41.7%).

Model 3: Gaussian Naive Bayes

Performance: F1 = 0.3620, Accuracy = 54.8%
Key finding: Independence assumption violated by correlated enrichment features; perfect precision (100%) but very low recall (22.1%).

Model 4: Linear Discriminant Analysis (LDA)

Performance: F1 = **0.7797**, Accuracy = 67.2% (best F1)
Key finding: Near-perfect recall (99.9%) makes it the best model for identifying premium rides.

Model 5: Perceptron

Performance: F1 = 0.5882, Accuracy = 58.5%
Key finding: Linear boundary struggles with this task but still benefits from enrichment (+16.2% F1).

7.3 Model Comparison

Table 6: Regression Model Comparison (Enriched Dataset)

Model	R^2	MAE
Decision Tree	**0.9696**	**$1.06**
Polynomial	0.9074	$1.90
Ridge	0.8963	$2.11
OLS	0.8962	$2.11
Lasso	0.8957	$2.10

Table 7: Classification Model Comparison (Enriched Dataset)

Model	F1	Accuracy	Precision	Recall
LDA	**0.7797**	67.2%	63.9%	99.9%
Logistic Reg	0.7494	67.6%	68.0%	83.5%
Decision Tree	0.7473	67.0%	67.2%	84.2%
Perceptron	0.5882	58.5%	69.3%	51.1%
Naive Bayes	0.3620	54.8%	100.0%	22.1%

7.4 Enrichment Impact

All 5 regression models improved. 4/5 classification models improved.

Table 8: Regression: Base vs Enriched (R^2)

Model	**Base R^2**	**Enriched R^2**	**Δ R^2**	**Improvement**
OLS	0.7013	0.8962	+0.1949	**+27.8%**
Ridge	0.7013	0.8963	+0.1950	**+27.8%**
Lasso	0.7005	0.8957	+0.1952	**+27.9%**
Decision Tree	0.9635	0.9696	+0.0061	+0.6%
Polynomial	0.8106	0.9074	+0.0969	**+12.0%**
Average	**0.7754**	**0.9130**	**+0.1376**	**+17.7%**

Table 9: Regression: Base vs Enriched (MAE)

Model	**Base MAE**	**Enriched MAE**	**Savings**
OLS	\$3.90	\$2.11	**-\$1.79**
Ridge	\$3.90	\$2.11	**-\$1.79**
Lasso	\$3.91	\$2.10	**-\$1.81**
Decision Tree	\$1.13	\$1.06	-\$0.07
Polynomial	\$3.14	\$1.90	**-\$1.24**

Table 10: Classification: Base vs Enriched (F1)

Model	**Base F1**	**Enriched F1**	**Δ F1**	**Improvement**
LDA	0.6817	0.7797	+0.0979	**+14.4%**
Perceptron	0.5064	0.5882	+0.0818	**+16.2%**
Decision Tree	0.6829	0.7473	+0.0644	**+9.4%**
Logistic Reg	0.6892	0.7494	+0.0602	**+8.7%**
Naive Bayes	0.6840	0.3620	-0.3220	-47.1%

Naive Bayes degraded because the enrichment features are correlated, violating its independence assumption. LDA achieved the largest gain (+14.4%), with near-perfect recall (99.9%) on the enriched dataset.

8. Solution: Ride Cost Optimization

8.1 Cost Function Design

To help riders minimize costs, we designed an optimization framework using our predictive models:

$$\text{Total Cost} = \text{Base Rate(service)} + (\text{Distance} \times \text{Per-Mile Rate}) \times \text{Surge Multiplier}$$

Our models reveal that the pricing formula can be decomposed into three components:

- **Service Selection:** Choosing UberX over Black saves ~$15 per ride on average
- **Surge Avoidance:** Waiting for surge to drop from 2.0× to 1.0× saves 50% of the fare
- **Platform Choice:** At equivalent service tiers, Uber and Lyft prices differ by ~$0.50–$1.00

8.2 Optimal Ride Recommendation

Using the Decision Tree model ($R^2 = 0.9696$), we can predict fares for different scenarios:
Scenario: 5-mile ride from Back Bay to Logan Airport.

Option	Service	Surge	Predicted Fare
Economy (no surge)	UberX	1.0×	$12.50
Economy (surge)	UberX	1.5×	$18.75
Premium (no surge)	Black	1.0×	$27.00
Platform switch	Lyft	1.0×	$11.80

Recommendation: The model recommends switching platforms (Uber ↔ Lyft), avoiding surge windows, and timing rides around weather conditions for the highest cost savings, projected to save $6.25–$15.20 per trip.

9. Summary and Conclusion

9.1 Summary of Outcomes

We developed a comprehensive white-box machine learning framework for analyzing 693K Uber and Lyft rides in Boston:

- **Regression:** Decision Tree achieved $R^2 = 0.9696$ (MAE = \$1.06)—predicting ride prices within \$1.06 on average
- **Classification:** LDA achieved F1 = 0.7797 with near-perfect recall (99.9%)
- **Enrichment:** Web-scraped data improved regression by **+17.7%** average R^2 and 4/5 classification models by up to **+16.2%** F1
- **Key Insight:** External enrichment features (weather severity, fuel indicators, transit patterns) capture pricing signals that base features alone miss, particularly for linear models

9.2 Practical Implications

The significant improvement from enrichment demonstrates that ride-hailing pricing is influenced by external conditions beyond the base features of distance, surge, and service type. Weather severity, fuel costs, and transit patterns all contribute additional predictive signal. This has implications for:

- **Consumers:** External conditions (weather, fuel prices) meaningfully affect ride costs—timing trips around these factors can reduce fares
- **Researchers:** Web-scraped enrichment from weather APIs and transit data provides genuine value for ride-hailing prediction
- **Platform operators:** The enriched models reduce prediction error by up to \$1.81 per ride for linear models—enabling better dynamic pricing

9.3 Future Work

1. **Real-Time Data Integration:** Incorporating live MBTA delay feeds and traffic congestion APIs to predict surge multiplier directly.
2. **Advanced Models:** Exploring XGBoost and Neural Networks for classification improvement.
3. **Deployment:** Building a Streamlit app for real-time fare estimation where users input pickup, dropoff, and time to receive instant predictions.
4. **Temporal Expansion:** Extending to multi-month datasets to capture seasonal pricing patterns.

10. References

1. Kaggle — Uber/Lyft Boston Rideshare Dataset. https://www.kaggle.com/datasets/brllrb/uber-and-lyft-dataset-boston-ma
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5. EIA — U.S. Energy Information Administration, regional gasoline price data.
6. Python Libraries: pandas, numpy, scikit-learn, statsmodels, matplotlib, scipy, requests.